

Benzodiazepines for depression? A review of the literature

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The English language literature on the use of benzodiazepines in depressive disorders was reviewed. We selected double-blind random assignment studies in which benzodiazepines were compared with reference drugs and/or placebo. Comparative studies with classical (non-triazolo) benzodiazepines in major depression show that these agents do not alleviate the core symptoms of depression, although they do have an effect on sleep and anxiety. Classical benzodiazepines show some efficacy in minor depression, but this conclusion could be related to efficacy in patients suffering from anxiety disorders rather than depression. Triazolo-benzodiazepines, mainly alprazolam, have been found to be effective in mild to moderate depression, although they turned out to be inferior to tricyclic antidepressants (TCAs) in patients with endogenous or melancholic depression. Furthermore it is questionable whether triazolo-benzodiazepines cause amelioration of the core symptoms of depression. With regard to combination therapy benzodiazepines may contribute to antidepressant response in the first weeks, because of a faster onset of effect than TCAs and/or because of effects on different symptoms. Beyond the first weeks of treatment combination therapy does not seem superior to monotherapy with TCAs.

Keywords: Classical benzodiazepines – Major depression – Minor depression – Symptom response – Triazolo-benzodiazepines – Tricyclic antidepressants

INTRODUCTION

Many patients are not treated with antidepressants, even when diagnosed as suffering from depression. In a recent study among general practitioners in the Netherlands approximately half of the patients diagnosed as depressed were treated with medication: 25% with tricyclic antidepressants (TCAs) and 75% with benzodiazepines (BZs; van den Brink *et al.*, 1991). These findings have been confirmed in other countries (Eisenberg, 1992). Although the use of BZs in the treatment of depression is regarded as less effective and consequently not as first choice treatment recent evidence indicates that at least alprazolam might be effective for major depression (Weissman *et al.*, 1992). We, therefore, examined the literature in order to try to answer the following questions:

- (1) What is the evidence for efficacy in depression of monotherapy with a BZ?
- (2) How effective are BZs in depression compared with TCAs?
- (3) Do BZs exert an effect on specific symptoms or subtypes of depression?
- (4) Is there a rationale for the combined use of TCAs and BZs in depression?

METHODS

The English language literature on the use of BZs in depressive disorders was reviewed, using the Medline program (CD-Rom) for 1968 through 1994, selecting double-blind, random assignment studies in which a BZ is compared with a TCA and/or placebo. Comparative studies with newer antidepressants would also be useful in assessing the possible antidepressant properties of benzodiazepines, because non-TCA antidepressants may have a somewhat different clinical profile. Only a few such studies have been conducted (Tiller *et al.*, 1989; Laws *et al.*, 1990), so a significant comparison cannot be made.

The triazolo-benzodiazepines (TBZs) are discussed separately, because in almost all recent studies a TBZ (usually alprazolam) is investigated on the basis of which they are claimed to have antidepressant properties.

RESULTS

Evaluation of the methods of the trials

All together 50 double-blind random assignment studies were found. Studies on classical benzodiazepines (CBZ) are listed in Table I, those on TBZ (or both) in Table II. The

TABLE Ia. Classical benzodiazepines: placebo-controlled studies with or without active comparator

Study	n	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Rickels <i>et al.</i> (1970)	243	Neurotic depression/HRSD and Zung	4 weeks AMI, max. 50 mg/day; COMB; PLA; CDZ max. 40 mg/day	AMI > CDZ in depression without anxiety. CDZ > AMI in anxiety without depression	AMI = CDZ = COMB ≥ PLA	No diagnostic criteria. No information on previous treatment. No information on drop-out rate
Claghorn <i>et al.</i> (1971)	120	Anxiety and depressive symptoms/BDI and Zung	4 weeks PRO 15 mg/day; OXA 45 mg/day; COMB; PLA	Not reported	PRO = OXA = COMB > PLA	No diagnostic criteria. No information on previous treatment. Very low dosage PRO
Covi <i>et al.</i> (1974)	146	Neurotic depression (♀)- POMS and SCL	16 weeks IMI 100–200 mg/day; DZP 10–20 mg/day; PLA	AMI > DZP on depression, obsession (SCL), irritability and anergia (POMS)	IMI > DZP = PLA	No diagnostic criteria. 34% previously treated with TCA. Exclusion: severe depression
Shammas (1977)	72	Anxiety or depressive neurosis/HRSD and BPRS	4 weeks AMI, mean 94 mg/day; BRO mean 24 mg/day; PLA	BRO > AMI on sleep, psychic and somatic anxiety	BRO > AMI = PLA	No diagnostic criteria. No information on previous treatment. 66% drop-outs in AMI sample. Low dosage AMI
Jacobson (1978)	95	Depression and anxiety, HRSD ≥ 20/HRSD and POMS	4 weeks AMI 75–100 mg/day; CDZ 30–60 mg/ day; COMB; PLA	Not reported	At 2 weeks: CDZ ≥ AMI 4 weeks: AMI > CDZ = COMB = PLA	No diagnostic criteria. No information on previous treatment. Low dosage AMI
Feighner <i>et al.</i> (1979)	279	Primary depression (Feighner criteria), HRSD > 20/HRSD and BDI	4 weeks AMI 75–150 mg/day; CDZ 1 30–60 mg/ day; COMB; PLA	AMI > CDZ on appetite, self- esteem, anergia, CDS > AMI on anxiety (HRSD)	At 2 weeks: CDZ > AMI At 4 weeks: COMB > AMI ≥ CDZ CDZ ≥ PLA	Dosage COMB > dosage AMI. Limited information on previous treatment. Fairly high drop-out rate (40%) in AMI sample
Johnstone <i>et al.</i> (1980)	240	Neurotic anxiety or depression/HRSD and HRSA	4 weeks AMI 150 mg/day; DZP 20 mg/day; COMB; PLA	Not reported	2 weeks: AMI = DZP 4 weeks: AMI > DZP AMI ≥ COMB, PLA ≥ DZP	No diagnostic criteria. Exclusion: recently treated patients and endogenous depressions
Lipman <i>et al.</i> (1986)	425	Primary depression (RDC), 75% major depression DSM- III/HRSD, HRSA, CAS and HSCL	8 weeks IMI, max. 200 mg/day; CDZ, max. 80 mg/day; PLA	IMI > CDZ on cognition, obsession, irritability. PLA > CDZ on irritability (HSCL)	1 week: CDS > IMI 3 weeks: CDZ + IMI 4 weeks: IMI > CDZ and PLA ≥ CDZ	No information on previous treatment. 33% drop-outs

AMI, amitriptyline; COMB, combination; PLA, placebo; CDZ, chloordiazepoxide; PRO, protriptyline; OXA, oxazepam; IMI, imipramine; DZP, diazepam; BRO, bromazepam.

> significantly more effective ($p < 0.005$); ≥ trend towards greater efficacy ($p < 0.1$); = no statistically significant difference.

* result is always measured with the main outcome criterion.

TABLE Ib. Classical benzodiazepines: studies with active comparator

Study	n	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Johnstone and Claghorn (1968)	30	Psychoneurosis with anxiety and depression/HRSA	6 weeks DOX mean 127 mg/day; CDZ mean 60 mg/ day	Not reported	DOX > CDZ	No diagnostic criteria. No information about previous treatment. Exclusion: affective psychosis
Verner (1969)	20	Reactive, neurotic secondary depression/MMPI	3 weeks IMI 75 mg/ day; CDZ 30 mg/ day	IMI > CDZ on mood, concentration, guilt, anxiety and anergia. IMI = CDZ on sleep	IMI > CDZ	No diagnostic criteria. Heterogenic patient sample. Low dosage IMI

continued ...

TABLE Ib. *Continued*

Study	n	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Rickels <i>et al.</i> (1969)	69	Anxiety nervosis and neurotic depression/HRSD and SCL	4 weeks DOX, mean 140 mg/day; DZP mean 11 mg/ day	DOX > DZP on retardation (HRSD) and mood (NSCL)	At 2 and 4 weeks DOX > DZP	No diagnostic criteria. No information about previous treatment. Fairly low dosages
Beaubien <i>et al.</i> (1970)	30	Psychoneurosis/ BPRS, WPRS and Zung	4 weeks DOX 75–150 mg/day; CDZ 30–60 mg/day	DOX > CDZ on mood and retardation. CDZ > DOX on anxiety (BPRS)	DOX = CDZ	No diagnostic criteria, > 50% drop-out in DOX treated sample. Dosage DOX fairly low
Sterlin <i>et al.</i> (1970)	30	Psychoneurotic reaction/BPRS, WPRS and Zung	4 weeks DOX 75–150 mg/day; CDZ 30–60 mg/day	DOX = CDZ on anxiety and obsessive symptoms (WPRS)	DOX = CDZ	No diagnostic criteria. 90% previously treated (with?). 50% drop-outs in DOX sample
Kay <i>et al.</i> (1970)	132	In-patients who underwent ECT/ HRSD, BDI	7 months AMI 50–150 mg/day; DZP 4–12 mg/day; both + ECT	Not reported	AMI > DZP	No diagnostic criteria. Hardly information on previous treatment. High drop-out rate
Montgomery <i>et al.</i> (1970)	41	Reactive depression, anxiety or both/HRSD	4 weeks DOX max. 90 mg/day; CDZ max. 90 mg/day	DOX > CDZ on mood, anxiety and cognition. CDZ > DOX on somatic symptoms	At 2 and 4 weeks: DOX > CDZ	No diagnostic criteria. No information on drop-outs and previous treatment
Sim <i>et al.</i> (1971)	65	Both anxiety and depressive symptoms/personal rating scale	6 weeks DOX 75–150 mg/day; CDZ 30–60 mg/day	DOX > CDZ on mood. CDZ > DOX on appetite, "neurotic behaviour"	DOX = CDZ	No diagnostic criteria. No information on previous treatment. Exclusion: more severely depressed patients
Wheatley (1971)	49	Neurotic depression/ personal	6 weeks AMI 75–150 mg/day; CDZ 30–60 mg/day	Not reported	At 1 week: AMI < CDZ, At 4 weeks AMI = CDZ, At 6 weeks: AMI > CDZ	No diagnostic criteria. No information on previous treatment. 44% experienced symptoms since 1–2 weeks
Sterlin <i>et al.</i> (1972)	50	Psychoneurosis/ BPRS and WPRS	4 weeks DOX 75–225 mg/day; CDZ 30–90 mg/day	DOX > CDZ on mood, anxiety, guilt (BPRS) and retardation (WPRS)	DOX > CDZ	No diagnostic criteria. Limited information on previous treatment. Low dosage AMI
Rickels <i>et al.</i> (1973)	93	Non-psychotic depression/HRSD and SCL	4–6 weeks AMI 75–125 mg/day; DZP 15–25 mg/day	AMI = DZP on sleep, anxiety and somatization (HRSD) AMI > DZP on depression, anxiety (SCL)	AMI ≥ DZP	No diagnostic criteria. Limited information on previous treatment. Low dosage AMI
Johnson <i>et al.</i> (1973)	100	Anxiety and depression/ personal	3 weeks DOT 75 mg/day; CDZ 30 mg/day	DOT > CDZ on anxiety, weight, depression, anergia	DOT > CDZ	No diagnostic criteria. Limited information on previous treatment. Low dosage DOT
d'Elia <i>et al.</i> (1974)	47	Anxiety or depression/ personal	8 weeks DOX, mean 55 mg/day; DZP mean 11 mg/ day	DOX > CDZ on sleep, depression, tension and anergia	At 1 week: DOX = CDZ 2–8 weeks: DOX > CDZ	No diagnostic criteria. No information on previous treatment. Exclusion: endogenous depression. Low dosage
Johnson (1979)	44	Neurotic depression, BDI > 11/HRSA and BDI	4 weeks AMI 75–150 mg/day; DZP 7.5–15 mg/day	Not reported	At 2 and 3 weeks: DZP > AMI. At 4 weeks: DZP = AMI	No diagnostic criteria. No information on previous treatment. Exclusion: endogenous depression. Dosages fairly low

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TABLE Ib. *Continued*

Study	<i>n</i>	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Lemoine <i>et al.</i> (1991)	118	Major depression without melancholia (DSM-III)/MADRS, HRSA, HSCL and CGI	4 weeks PRA 30–60 mg/day; CLO 75–150 mg/ day	Not reported	CLO = PRA. More severely ill subgroup: CLO > PRA	Exclusion: melancholia, suicidality. Number of responders not reported. No information on previous treatment. Fairly low dosage CLO

DOX, doxepin; DOT, dothiepin; PRA, prazepam; CLO, clomipramine; see Table Ia for other abbreviations.

> significantly more effective ($p < 0.005$); $\geq$ trend towards greater efficacy ($p < 0.1$); = no statistically significant difference.

* result is always measured with the main outcome criterion.

TABLE IIa. Triazolo-benzodiazepines: placebo-controlled studies with or without active comparator

Study	<i>n</i>	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Feighner <i>et al.</i> (1983)	723	Primary depression (Feighner criteria), HRSD $\geq$ 18/HRSD, HRSA and HSCL	6 weeks IMI, mean 131 mg/day; ALP, mean 2.9 mg/day; PLA	ALP = IMI on mood, retardation, weight, cognition. ALP > IMI on sleep and anxiety (HRSD)	At weeks 1 and 2 = ALP > IMI At week 6: IMI = ALP > PLA	No information about previous treatment. 54% responders in PLA sample. Exclusion: psychomotor retardation, bipolar patients
Imlah (1985)	65	Neurotic depression/HRSA, CAS, Raskin and CGI	4 weeks AMI max. 150 mg/day; ALP max. 3 mg/day; PLA	Not reported	ALP $\geq$ AMI ALP > PLA	Low dosage AMI, no diagnostic criteria, 10 information on previous treatment
Rickels <i>et al.</i> (1985)	504	Primary depression (Feighner criteria), HRSD $\geq$ 18/HRSD, HRSA and HSCL	6 weeks AMI max. 225 mg/day; DOX max. 225 mg/day; ALP max. 3.0 mg/ day; PLA	ALP = DOX = AMI on anxiety, retardation, sleep (HRSD). AMI = DOX > ALP on mood (HSCL)	At week 1: ALP > AMI = DOX At 6 weeks: ALP = AMI = DOX > PLA	No information on previous treatment. Fairly low response rate on TCA (50%). Exclusion: bipolarity, involutional depression
Smith <i>et al.</i> (1986)	80	Major depression DSM-III/HRSD, CGI, HSCL and Carroll. In-patients	6 weeks ADI, mean 46 mg/day; PLA	ADI had no specific effects on different clusters	At weeks 1, 2, 4 and 6: ADI > PLA	Hardly information on previous treatment. Exclusion criteria not reported. High drop-out rate (50%)
Mendels and Schless (1986)	98	HRSD $\geq$ 20/HRSD and HRSA	6 weeks IMI, mean 167 mg/day; ALP mean 3.7 mg/day; PLA	Not reported	Weeks 1–4: ALP = IMI Weeks 5–6: ALP > IMI, ALP $\geq$ PLA	No diagnostic criteria. No information on previous treatment. 38% responders on IMI. Drop-out rate 50%
Dunner <i>et al.</i> (1987)	80	Major depression DSM-III, 31% bipolar II patients/ HRSD, HSCL, CGI and Carroll	6 weeks ADI 20–90 mg/day; PLA	ADI > PLA on sleep and anxiety	ADI > PLA	No information about previous treatment. Exclusion criteria not reported. High drop-out rate (50%)
Fawcett <i>et al.</i> (1987)	79	Major depression DSM-III HRSD $\geq$ 20/HRSD, HRSA, Zung and BDI	6 weeks DMI mean 230 mg/day; ALP mean 4–6 mg/day; COMB; PLA	Not reported	1 week: ALP > DMI = COMB = PLA 2–6 weeks: ALP = DMI = COMB > PLA	No information on previous treatment. Exclusion: suicidality, psychotic depression. % responders not reported
Rickels <i>et al.</i> (1987)	241	Major depression DSM-III, 64% with melancholia, HRSD $\geq$ 18/HRSD, CGI and HSCL	6 weeks IMI mean 143 mg/day; ALP mean 3.1 mg/day; DZP mean 24 mg/ day; PLA	IMI = ALP on anxiety, retardation, cognition ALP > IMI on sleep (HRSD)	1 week: ALP > IMI = DZP = PLA 4–6 weeks: IMI > ALP > DZP = PBO	Limited information on previous treatment. 39% drop-outs. Exclusion: bipolarity, psychosis

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TABLE IIa. *Continued*

Study	n	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Cohn <i>et al.</i> (1988)	72	Major depression DSM-III/HRSD, CGI, HSCL and Carroll	6 weeks ADI 45–90 mg/day; PLA	ADI > PLA on sleep, anxiety and retardation	ADI > PLA	Hardly information on previous treatment. 60% drop-outs. Exclusion criteria not reported
Hicks <i>et al.</i> (1988)	48	Major depression with melancholia, HRSD $\geq$ 26/HRSD, CGI	6 weeks AMI mean 142 mg/day; ADI mean 50 mg/day; PLA	Not reported	1 week: AMI < ADI 2–6 weeks: AMI > ADI = PLA	No information on previous treatment. % responders not reported
Borison <i>et al.</i> (1989)	119	Major depression DSM-III HRSD $\geq$ 18/HRSD	4–6 weeks ALP 6 mg/day; ALP 3 mg/day; PLA	Not reported	ALP 6 = ALP 3 = PLA	No information on previous treatment, drop-out rate and % responders
Rickels <i>et al.</i> (1991)	259	Major depression DSM-III/HRSD, CGI, HSCL and CAS	6 weeks IMI mean 159 mg/day; ADI mean 65 mg/day; DZP mean 35 mg/ day; PLA	IMI > ADI and DZP on anxiety and retardation. ADI and DZP > IMI on sleep disturbance	IMI > ADI > DZP = PLA	Exclusion: suicidality. Fairly high drop-out rate (43%). Little information on previous treatment
Weissman <i>et al.</i> (1992)	35	Major depression DSM-III/HRSD, CAS, MMSE	6 weeks IMI mean 97.5 mg/day; ALP mean 2.2 mg/day; PLA each + IPT	ALP > IMI on sleep ALP = IMI on retardation and anxiety	1–2 weeks: ALP > IMI = PLA 6 weeks: ALP = IMI = PLA	43% previously treated (with?); 40% drop-outs. Low dosage and low response rate on IMI. Elderly patients

ALP, alprazolam; ADI, adinazolam; DMI, desipramine; see Table I for other abbreviations.

> significantly more effective ($p < 0.005$); $\geq$ trend towards greater efficacy ($p < 0.1$); = no statistically significant difference.

* result is always measured with the main outcome criterion.

TABLE IIb. Triazolo-benzodiazepines: studies with active comparator

Study	n	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Ansseau <i>et al.</i> (1984)	126	Primary non- psychotic depression/HRSD, HRSA, CGI and BDI	6 weeks DOX mean 137 mg/day; ALP mean 2.7 mg/ day	Not reported	At weeks 1, 2, 4 and 6: DOX = ALP	No diagnostic criteria. Number of responders not reported. No information on previous treatment and drop-outs. Exclusion: bipolarity, psychotic features
Lenox <i>et al.</i> (1984)	21	Major depression (RDC) in-patients/ HRSD, PGI and SADS	24 days IMI mean 243 mg/day; ALP mean 5.2 mg/day; ECT 10–15 treatments	Not reported	2–3 weeks: ECT = IMI > ALP	No information on previous treatment. Exclusion criteria not reported. Short duration of trial
Remick <i>et al.</i> (1985)	54	Major depression (RDC). 69% had shortened REM latency. 40% in-patients/HRSD and CGI	6 weeks DMI mean 208 mg/day; ALP mean 3.8 mg/day	Not reported	Out-patients: DMI = ALP In-patients: DMI $\geq$ ALP	No information on previous treatment. Low plasma DMI levels. Out-patients: low response rate (36%)
Rush <i>et al.</i> (1985)	52	Major depression RDC, HRSD $\geq$ 18, shortened REM latency/HRSD, SCL, BDI and CAS	6 weeks AMI mean 190 mg/day; ALP mean 4.4 mg/day	AMI > AIP on mood, psychoticism and irritability (SCL)	Weeks 1–2: ALP > AMI Weeks 4–6: AMI > ALP	No information on previous treatment. Limited information on drop-outs. 35% in-patients
Amsterdam <i>et al.</i> (1986)	40	Major depression DSM-III/HRSD and CGI	12 weeks IMI mean 148 mg/day; ADI mean 70 mg/day	Not reported	Weeks 1–5: IMI = ADI Weeks 6–12: IMI $\geq$ ADI	No information on previous treatment. 100% responders on IMI, 50% drop-outs

continued ...

TABLE IIb. *Continued*

Study	<i>n</i>	Diagnosis/ assessment	Duration and dosage	Symptom response	Result	Remarks
Goldberg <i>et al.</i> (1986)	60	Major depression with melancholia DSM-III, HRSD $\geq$ 18/HRSD, HRSA and HSCL	6 weeks IMI mean 193 mg/day; ALP 3.9 mg/day	IMI > ALP on mood (HRSD), obsession and psychoticism (HSCL)	At weeks 1–2: ALP > IMI Weeks 4–6: IMI > ALP	No information on previous treatment. Response rate not reported. Exclusion: bipolarity
Overall <i>et al.</i> (1987)	90	Major depression DSM-III, Zung $\geq$ 55/HRSD and BPRS	6 weeks IMI mean 200 mg/day; ALP mean 3.6 mg/day	Not reported	2 weeks: ALP > IMI 4–6 weeks: ALP = IMI	No information on previous treatment. % responders not reported. 43% drop-outs on IMI
Eriksson <i>et al.</i> (1987)	81	Major depression DSM-III, HRSD $\geq$ 18/HRSD, HRSA and CGI	6 weeks AMI mean 150 mg/day; ALP mean 3.0 mg/day	Not reported	1–2 weeks: AMI = ALP 4–6 weeks: AMI > ALP	Dosage AMI fairly low. Exclusion: episode duration > 2 years. 94% responders on AMI
Singh <i>et al.</i> (1988)	130	Moderate depression, HRSD $\geq$ 18/HRSD, HRSA, CGI and HSCL	6 weeks AMI mean 135 mg/day; ALP mean 2.4 mg/day	ALP > AMI on anxiety (HRSD)	1–3 weeks: ALP > AMI 4–6 weeks: ALP = AMI	No diagnostic criteria. No information on previous treatment. Exclusion: psychotic depression, bipolarity
Remick <i>et al.</i> (1988)	52	Major depression RDC/HRSD $\geq$ 21/HRSD and CGI	6 weeks DMI mean 192 mg/day; ALP mean 3.3 mg/day	Not reported	2 weeks: ALP > DMI 4–6 weeks: ALP = DMI	No information on previous treatment. DMI sample: more chronic depressives. Low response rate on DMI (42%). Exclusion: bipolarity
Hubain <i>et al.</i> (1990)	30	RDC major depression, endogenous acc. to Newcastle I in- patients, HRSD $\geq$ 25/HRSD, HRSA and Zung	6 weeks AMI mean 167 mg/day; ALP mean 8.5 mg/day	Not reported	1–2 weeks: AMI = ALP 3–6 weeks: AMI > ALP	No information on previous treatment and % responders. 50% drop-outs
Kennedy <i>et al.</i> (1991)	31	Major depression DSM-III-R/HRSD, HRSA, CGI, BDI and HSCL (14 in-patients)	6 weeks DMI mean 202 mg/day; ADI mean 81 mg/day	Not reported	Weeks 1, 2, 4 and 6: DMI = ADI	No information on previous treatment. Fairly low response rate (40%). No exclusion criteria reported
Ansseau <i>et al.</i> (1991)	66	Major depression RDC with DST non-suppression or abnormal CNV, in-patients/HRSD and CGI	4 weeks AMI mean 207 mg/day; ADI mean 87 mg/day; DZP mean 41 mg/ day	Not reported	AMI > DZP AMI $\geq$ ADI	No information on previous treatment. No exclusion criteria reported
Lapierre <i>et al.</i> (1994)	43	Moderate depression/HRSD, HRSA, BDI, CGI and HSCL	6 weeks AMI mean 125 mg/day; ALP mean 3.2 mg/day	AMI > ALP on anxiety, cognition and suicidality	1 week: AMI = ALP 3–6 weeks: AMI > ALP	No diagnostic criteria. Exclusion: bipolarity, psychotic depression. No information on previous treatment. Fairly low dosage AMI

See Tables I and IIa for abbreviations.

> significantly more effective ($p < 0.005$); $\geq$ trend towards greater efficacy ($p < 0.1$); = no statistically significant difference.

* result is always measured with the main outcome criterion.

following items are considered important in the interpretation of study results: diagnostic criteria employed, patient selection, previous treatment, placebo control, dosage and duration of pharmacotherapy and drop-out rate. We explored both whether information about the specific items was presented and, if possible, the validity of that information.

Diagnostic criteria and patient selection. In 26 trials a classification system was employed, either DSM-III-R, Research Diagnostic Criteria (RDC) or Feighner criteria for depression. Paykel *et al.* (1988) and Stewart *et al.* (1983) have found that patients with major depression according to these criteria are to be expected to benefit from TCAs as opposed to those meeting criteria for minor

depression according to RDC criteria. We, therefore, made the same distinction in this review.

With regard to those trials not employing a classification system, we tried to classify the study population as representing patients with major depression or minor depression, on the basis of the description in the methodology section of the paper. Study populations not meeting the criteria for major depression were classified as minor depression. There was some doubt about a few study populations, which were classified as major depression (Covi *et al.*, 1974; Lipman *et al.*, 1986; Lapierre *et al.*, 1994). With regard to many studies performed before 1980 and one more recent study (Imlah, 1985) it turned out to be difficult to draw the line between minor depression and anxiety disorders. Studies in which none of the above classification systems was employed are, therefore, judged less valid. Despite the use of the DSM-III-R classification "major depression" and of the Hamilton Rating Scale for Depression (HRSD) or the Montgomery-Åsberg Depression Rating Scale (MADRS) in nearly all of the more recent studies, there are considerable differences between study populations. The way subjects are recruited is a factor determining the characteristics of the study population and possibly also treatment effects (Ansseau, 1992). A few papers mention recruitment by means of newspaper advertisements (Lipman *et al.*, 1986; Goldberg *et al.*, 1986; Dunner *et al.*, 1987; Hicks *et al.*, 1988). This may result in populations with characteristics different from populations selected at out-patient clinics (Amori and Lenox, 1989). Other studies do not mention the method of patient recruitment. Also considered significant is the exclusion in 18 studies of patients with an "endogenous" depression, bipolar or severe depression, because this may bias selection towards mixed anxiety-depression or less severe syndromes (Johnstone and Claghorn, 1968; Sim *et al.*, 1971; d'Elia *et al.*, 1974; Covi *et al.*, 1974; Johnson, 1979; Johnstone *et al.*, 1980; Feighner *et al.*, 1983; Ansseau *et al.*, 1984; Rickels *et al.*, 1985, 1987, 1988, 1991; Fawcett *et al.*, 1985; Imlah, 1985; Eriksson *et al.*, 1987; Singh *et al.*, 1988; Lemoine *et al.*, 1991; Lapierre *et al.*, 1994).

Previous treatment. In major depression, previous unsuccessful treatment with a TCA during the index episode reduces the response rate of a second TCA (Nolen and Haffmans, 1989). Since the inclusion of TCA-resistant patients may affect outcome, information regarding previous pharmacotherapy for the index episode may be important. Only two studies present clear information concerning previous treatment with TCA (Covi *et al.*, 1974; Johnstone *et al.*, 1980). In nine other trials information regarding the number of patients previously treated is given, but neither the drug nor the duration of treatment is mentioned (Sterlin *et al.*, 1970; Rickels *et al.*, 1973, 1987,

1991; Jacobson, 1978; Smith and Glaudin, 1986; Cohn *et al.*, 1988; Weissman *et al.*, 1992).

Placebo control. A placebo-controlled parallel group study is the design of choice to test the efficacy of a drug in depression. Rabkin *et al.* (1990) refer to expectancy effects, spontaneous improvement, therapeutic benefits of repeated assessments and response bias as aspecific treatment factors in the kind of trials evaluated here. In some out-patient studies (Feighner *et al.*, 1983) a remarkably high percentage of placebo responders is found, indicating a large effect of these aspecific treatment factors. A high percentage of placebo response may be related to the selection of out-patients, patients with mild depression, patients with non-endogenous depression and symptomatic volunteers (Angst *et al.*, 1989). Twenty-one studies include a placebo-treated group (Rickels *et al.*, 1970, 1985, 1987, 1991; Claghorn and Kellner, 1971; Covi *et al.*, 1974; Shamma, 1977; Jacobson, 1978; Feighner *et al.*, 1979, 1983; Johnstone *et al.*, 1980; Imlah, 1985; Mendels and Schless, 1986; Lipman *et al.*, 1986; Smith and Glaudin, 1986; Fawcett *et al.*, 1987; Dunner *et al.*, 1987; Cohn *et al.*, 1988; Hicks *et al.*, 1988; Borison *et al.*, 1989; Weissman *et al.*, 1992).

Dosage and duration of treatment. One of the problems in trials comparing two drugs is to define optimal/adequate dosage and duration of treatment for each of them. Treatment with TCAs is considered adequate if a patient has received the equivalent of at least 150 mg imipramine during at least 4 weeks (Fawcett and Kravitz, 1985), although some patients may need higher doses to achieve therapeutic plasma levels (Quitkin, 1985) and/or extension of the treatment time (Quitkin *et al.*, 1984). Only 17 out of 50 comparative studies met these minimum standards for adequate treatment with a TCA (Sterlin *et al.*, 1972; Covi *et al.*, 1974; Johnstone *et al.*, 1980; Rickels *et al.*, 1985, 1991; Remick *et al.*, 1985, 1988; Rush *et al.*, 1985; Lipman *et al.*, 1986; Amsterdam *et al.*, 1986; Mendels and Schless, 1986; Goldberg *et al.*, 1986; Fawcett *et al.*, 1987; Overall *et al.*, 1987; Hubain *et al.*, 1990; Kennedy *et al.*, 1991; Ansseau *et al.*, 1991). Thus, inadequate treatment may have obscured the superiority of the TCAs over the BZs in other studies. Another problem is that it is less clear what should be considered an adequate dose and duration in the treatment with a BZ in depression.

Drop-out rate. Placebo-treated patients often drop out of a study because of lack of improvement, while in patients receiving active compounds side effects often are an important cause of drop out. Patients who experience an early recovery may also be inclined to premature termination of drug therapy. Drop out or withdrawal from clinical trials is a serious problem for interpretation of the

results, particularly when there is a substantial difference between the treatment groups regarding the drop-out rate or the reasons of drop out (Angst *et al.*, 1989). In general a drop-out rate up to 30% is considered acceptable (Angst *et al.*, 1989). In five reports no information is given regarding the drop-out rate (Rickels *et al.*, 1970; Montgomery *et al.*, 1970; Ansseau *et al.*, 1984; Lenox *et al.*, 1984; Borison *et al.*, 1989). In six studies a high drop-out rate is limited to the TCA-treated patients, largely caused by adverse effects (Beaubien *et al.*, 1970; Sterlin *et al.*, 1970; Rickels *et al.*, 1973; Shammass, 1977; Feighner *et al.*, 1979; Overall *et al.*, 1987). In six other studies a high overall drop-out rate is reported, the reasons being different between treatments; in five of these studies drop out is caused mainly (Kay *et al.*, 1970; Hubain *et al.*, 1990) or partially (Amsterdam *et al.*, 1986; Rickels *et al.*, 1987, 1991) by inefficacy in BZ-treated patients. In TCA-treated subjects drop out is mainly caused by adverse effects, except in two studies (Mendels and Schless, 1986; Weissman *et al.*, 1992) when lack of improvement is also a major reason for drop out. In three studies the high number of drop outs is confined to the placebo group and largely caused by inefficacy (Smith and Glaudin, 1986; Dunner *et al.*, 1987; Cohn *et al.*, 1988). The drop-out rate of the remaining 29 studies appears to be acceptable.

Summary of the methodology. When taking together all the items discussed, only nine studies fulfil most of the criteria although none fulfils all. The shortcomings of these nine most valuable studies are as follows. In the Covi *et al.* (1974) study diagnostic criteria are lacking and severely depressed patients are excluded. The study of Rush *et al.* (1985) lacks a placebo-treated group. In the Rickels *et al.* (1985) study severely depressed patients are excluded. Goldberg *et al.* (1986) and Fawcett *et al.* (1987) do not report on the number of responders and exclude severely depressed patients and Goldberg *et al.*'s study lacks a placebo group. Kennedy *et al.* (1991) and Ansseau *et al.* (1991) do not report exclusion criteria, both studies lack a placebo group and the validity of the Kennedy *et al.* study can be questioned because of a low response rate in the desipramine-treated group. In the Rickels *et al.* (1991) study severely depressed patients are excluded and the overall drop-out rate is high. All except the Covi *et al.* (1974) study lack valuable information about previous treatment with TCAs.

General evaluation of efficacy

To assess the antidepressant efficacy of BZs emphasis is put on the comparative studies in major depression, this being the main indication for treatment with antidepressants.

CBZ in major depression. In none of the five studies comparing a CBZ with placebo was a statistically significant

difference found. All of these studies were three or four leg studies, including a TCA-treated group, the TCA being superior over placebo in every study. Three of the five studies belonged to the methodologically most valid ones. These trials were quite large, the total number of patients ranging from 146 (Covi *et al.*, 1974) to 425 (Lipman *et al.*, 1986). It therefore seems unlikely that a clinically relevant difference in efficacy was missed because of an insufficient number of patients. In three other studies a CBZ was compared with a TCA in patients with major depression. One of the studies (Anseau *et al.*, 1991) fulfilled most of the criteria mentioned earlier. Two of these found the TCA to be superior, while in the remaining study (Lemoine *et al.*, 1991) the TCA was superior only in a subgroup of severely ill patients.

Thus, the conclusion can be drawn that there is no evidence of an antidepressant effect for the CBZs in major depression. They did not show any advantage over placebo and were clearly inferior to TCAs. In most studies the dosages of the CBZs were those used for the treatment of anxiety. Whether higher dosages would have been more beneficial is unknown.

TBZ in major depression. In 11 studies a TBZ was compared with placebo. In eight of them (Feighner *et al.*, 1983; Rickels *et al.*, 1985, 1987, 1991; Smith *et al.*, 1986; Dunner *et al.*, 1987; Fawcett *et al.*, 1987; Cohn *et al.*, 1988) the TBZ was significantly more effective, while in two trials no statistically significant difference was found. The three studies considered methodologically most valid (Rickels *et al.*, 1985, 1991; Fawcett *et al.*, 1987) all reported the TBZ to be superior over placebo. These studies were fairly large three leg studies, the total number of patients ranging from 79 (Fawcett *et al.*, 1987) to 504 (Rickels *et al.*, 1985). In two of them no difference was found between the TBZ and the TCA; the remaining study reported the TCA to be superior over both the TBZ and placebo (Rickels *et al.*, 1991).

In the majority of these studies patients were required to have a minimum baseline 21-item HAM-D score of 18, which is below the minimum level of 18 on the 17-item version usually applied in out-patient antidepressant trials, indicating that more mildly depressed patients may have been included.

In 19 studies a TBZ was compared with a TCA in patients with major depression. In 10 studies the TCA showed to be superior, while in the remaining nine studies no difference was found. In seven of these studies special patient groups were selected. The TCAs turned out to be superior in patients with severe major depression with melancholia (Goldberg *et al.*, 1986; Rickels *et al.*, 1987), in patients with endogenous depression according to the Newcastle I Scale (Hubain *et al.*, 1990), in patients selected on the basis of a shortened REM latency (Rush *et al.*, 1985), or selected on the basis of an abnormal DST (Ans-

seau *et al.*, 1991) and in in-patients (Lenox *et al.*, 1984; Remick *et al.*, 1985). In the majority of the remaining 10 studies, patients with bipolar depression, melancholic subtype or suicidality were excluded. Apparently, most subjects in these studies suffered from major depression without melancholia. In only three out of these 11 studies (Eriksson *et al.*, 1987; Rickels *et al.*, 1991; Lapierre *et al.*, 1994) was the TCA identified as superior; the other studies failed to show a statistically significant difference between TBZs and TCAs.

It is concluded that antidepressant efficacy of TBZ over placebo has been proven. However, TBZs are inferior to TCAs in severe or endogenous depression, while efficacy equal to TCAs appears to be restricted to patients with major depression without melancholia. In most studies the dose of TBZ was approximately twice the recommended dose for anxiolytic effects.

TBZs versus CBZs in major depression. In three studies TBZs were compared with CBZs in patients with major depression. Two of these studies belong to the methodologically most valid ones (Rickels *et al.*, 1991; Ansseau *et al.*, 1991). In the two largest studies, which included a total of 241 (Rickels *et al.*, 1987) and 259 (Rickels *et al.*, 1991) patients, respectively, the TBZ was superior over the CBZ. Both studies included a placebo-treated group, the CBZ showing no difference in efficacy with placebo. In the remaining study (Ansseau *et al.*, 1991), which included a total of 66 patients, no significant difference in efficacy was found between the TBZ and the CBZ.

Studies in minor depression. With regard to those trials not employing a classification system, we tried to classify the study population as representing patients with major depression or minor depression. Populations not meeting the criteria for major depression were classified as minor depression. In most of these studies subjects carry a diagnosis of neurotic depressive reaction, psychoneurotic reaction with depression or anxious depression, so the distinction with anxiety disorders seems hard to make. Out of five studies comparing a CBZ with placebo, the CBZ showed superiority in two. In 18 studies a CBZ was compared with a TCA. Ten reported the TCA to be superior, one identified the CBZ as superior and seven studies did not show a statistically significant difference. With regard to TBZ only four studies in minor depression were found, so it does not seem useful to draw any conclusion concerning their efficacy. Most studies in minor depression were carried out before 1980 and, besides the lack of classification, they show several other methodological flaws. In conclusion, CBZs appear to have some antidepressant effect in minor depression, although they are inferior to TCAs. More specific conclusions cannot be drawn.

Symptom response

The usual instruments to assess the severity of depression are the HRSD and the MADRS. These scales do not only assess mood and melancholic symptoms, but other symptoms like anxiety, agitation and sleep disturbances as well. Thus, drugs that do not influence the core symptoms of depression, namely anhedonia, psychomotor retardation and anergia (Schatzberg and Cole, 1978; Johnson, 1985), can still show an effect by alleviating insomnia or anxiety. It is questionable whether this should be called an antidepressant effect. In order to solve this problem it may be helpful to look at the response of symptom clusters or of individual symptoms. Although assessing "true" antidepressant efficacy by looking at the response of individual symptoms could be promising, only about half of the studies used this approach. Most report on the effect of the drugs on depressive mood, but there is substantial variation in reporting of other symptoms. In some studies it remains obscure whether a specific symptom was examined or whether no difference was found between the two drugs in alleviating this symptom. Because symptom response may differ between diagnostic groups, comparative studies reporting on this topic were divided into three categories: studies with CBZs in major depression, studies with TBZs in major depression and studies with CBZs in minor depression. Studies involving both a CBZ and a TBZ are discussed in both sections.

CBZs in major depression. Only five studies comparing a CBZ with a TCA in major depression report on the symptom response. In two of these studies the TCA had a superior effect on depressive mood (Covi *et al.*, 1974; Lipman *et al.*, 1986); in two other studies the TCA had more effect on retardation (Rickels *et al.*, 1987, 1991). Covi *et al.* (1974) and Feighner *et al.* (1979) found the TCA to have more effect on anergia. With regard to the effect of the drugs on anxiety the TCAs were superior in three studies (Lipman *et al.*, 1986; Rickels *et al.*, 1987, 1991), while CBZs appeared superior in one study (Feighner *et al.*, 1979). CBZs had a superior effect on sleep disturbances in one study (Rickels *et al.*, 1991), while in two studies no difference was found (Feighner *et al.*, 1979; Rickels *et al.*, 1987). We conclude, therefore, that TCAs are superior to CBZs in alleviating the "core" symptoms of depression. TCAs have a superior effect on depressive mood, but with respect to retardation and anergia no clear differences emerge.

TBZs in major depression. Eight studies comparing a TBZ with a TCA in major depression report on the symptom response. In four of them the TCA had a superior effect on depressive mood (Rickels *et al.*, 1985; Rush *et al.*, 1985; Goldberg *et al.*, 1986; Lapierre *et al.*, 1994), while in one study no difference was found (Feighner *et al.*

al., 1983). In one study the TCA had more effect on retardation (Rickels *et al.*, 1991), while in four studies no difference was found (Feighner *et al.*, 1983; Rickels *et al.*, 1985, 1987; Weissman *et al.*, 1992). TBZs appeared to have more effect on sleep disturbances in four studies (Feighner *et al.*, 1983; Rickels *et al.*, 1987, 1991; Weissman *et al.*, 1992). The effect of TBZs on anxiety was superior in one study (Feighner *et al.*, 1983), inferior in another (Rickels *et al.*, 1991), while in three trials no difference was found (Rickels *et al.*, 1985, 1987; Weissman *et al.*, 1992) and in one study the TCA was more effective (Lapierre *et al.*, 1994).

Eight placebo-controlled studies with TBZs report on the symptom response. TBZs had a superior effect on sleep in seven studies, while in one study no difference was found.

In six studies the TBZ was superior to placebo on anxiety; in two trials no difference was found. With regard to the effect on psychomotor retardation, the TBZs were superior in four studies (Feighner *et al.*, 1983; Rickels *et al.*, 1985, 1987; Cohn *et al.*, 1988), while in another four studies no difference was found (Smith *et al.*, 1986; Dunner *et al.*, 1987; Rickels *et al.*, 1991; Weissman *et al.*, 1992). None of these studies reported on the effect on anergia. With regard to weight loss and diurnal variation, which are sometimes considered to belong to the "core" symptoms, no difference was found in all four studies reporting on weight loss (Feighner *et al.*, 1983; Dunner *et al.*, 1987; Cohn *et al.*, 1988; Weissman *et al.*, 1992), as well as in the three studies reporting on diurnality (Feighner *et al.*, 1983; Dunner *et al.*, 1987; Weissman *et al.*, 1992).

In five studies different subscales of the HAM-D were analysed. In the Rickels *et al.* (1987) study the TBZ was superior over placebo according to both the Rickels "core cluster" (Rickels *et al.*, 1987) and the Bech "depression cluster" (Bech *et al.*, 1975).

In two other studies (Dunner *et al.*, 1987; Rickels *et al.*, 1991) TBZs proved to be superior over placebo according to the "endogenous" scale (Thase *et al.*, 1983). In the Kennedy *et al.* (1991) study no difference was found between the TBZ and the TCA according to both the "depression cluster" and the "endogenous" scale. TCAs were superior over TBZs in two studies (Rickels *et al.*, 1991; Ansseau *et al.*, 1991) according to the "endogenous" scale and in one study (Rickels *et al.*, 1991) according to the Rickels "core cluster". Although the analysis of these subscales might be a useful way of assessing "true" antidepressant effect, only in a few studies were these subscales actually analysed and the results are ambiguous. In conclusion, TCAs possibly are superior to TBZs with regard to the "core" symptoms of depression, although this is somewhat less clear than with CBZs. When compared with placebo, TBZs seem to have a superior effect on psychomotor retardation, but not on other "core" symptoms. TBZs clearly are superior over placebo regarding their effect on anxiety and sleep.

CBZs in minor depression. Eleven studies comparing a TCA with a CBZ in minor depression report on the symptom response. Eight studies report on the effect on depressive mood, TCAs being superior in every one.

TCAs appear to be superior in all three studies reporting on retardation, as well as in the two studies reporting on anergia. With regard to anxiety TCAs were found to be superior to CBZs in four studies, inferior in one, while in another study no difference was found. CBZs had more effect on sleep disturbances in one study, TCAs were superior in another study, while in one study no differences emerged.

Onset and maintenance of effect

Thirty-one studies provide data on measurement between baseline and endpoint. In 16 studies the BZs were found superior at Week 1 or 2 and TCAs in only two. In contrast, at Week 3 or 4, however, TCAs were found superior in 14 studies, while in only one was the BZ superior. In summary, the effect of BZs often appears during the first 2 weeks, while it takes 3–4 weeks for a TCA to achieve efficacy. Concerning the onset of action, there seems to be no distinction between CBZs and TBZs.

With regard to TBZs in most studies a maximum effect is achieved after 2 weeks, while a few studies mention an increasing effect during the 6 weeks study period. In almost all studies reporting a maximum effect at Week 2, the efficacy is sustained during the study period; only one study (Rickels *et al.*, 1991) reported a decreasing effect of the TBZ. These findings may be biased, because of patients dropping out of the studies, due to inefficacy. The percentage of patients on TBZ dropping out due to lack of effect is not always reported, but it amounts to 25% on average. On the other hand, the efficacy of TCAs usually increases during the study period, while on average less TCA-treated patients drop out due to lack of effect. Thus, the onset of action of TBZs appears to be swifter than with TCAs. As to maintenance of effect, it cannot be excluded that efficacy of TBZs decreases over weeks. However, firm evidence is lacking. Moreover, studies do not extend beyond 6 weeks. Efficacy of TBZs or CBZs compared with TCAs over periods longer than 6 weeks is an important issue, but no controlled studies are available.

Combination of BZs and TCAs

In clinical practice, TCAs and BZs are often used in combination, probably because of differences in symptom responding and onset of action. The question is whether there is an advantage in favour of combination therapy over treatment with a TCA or a BZ alone. Nine studies compared combination therapy with TCA alone or TCA plus placebo (Table III). In only four trials were diagnostic criteria employed (Bowen, 1978; Feighner *et al.*, 1979; Feet *et al.*, 1985; Fawcett *et al.*, 1987). Information con-

TABLE III. Combination therapy

Study	n	Diagnosis	Duration and dosage	Result	Remarks
Rickels <i>et al.</i> (1970)	24 3	Neurotic depression	4 weeks AMI max. 100 mg/day; combination with CDZ max. 40 mg/day	COMB > AMI in patients with high anxiety. COMB = AMI in patients with low anxiety	No diagnostic criteria. No information on previous therapy and drop-outs. Unclear presentation of results
Claghorn <i>et al.</i> (1971)	12 0	Anxiety and depressive symptoms	4 weeks PRO 15 mg/day; combination with OXA 45 mg/day	PRO = COMB	No diagnostic criteria. Low dosage PRA. No information on previous treatment
Hare (1971)	40	Psychoneurotic reaction	5–6 weeks AMI 50– 112.5 mg/day; combination with CDZ 20–45 mg/day	At 2 weeks: COMB > AMI Endpoint: COMB = AMI	No diagnostic criteria. No information on previous treatment and drop-outs. Low dosage AMI
Jacobson (1978)	95	Depression and anxiety, HRSD $\geq$ 20	4 weeks AMI 75–100 mg/ day; combination with CDZ 30–60 mg/day	At 3 weeks: COMB = AMI At 4 weeks: AMI $\geq$ COMB	No diagnostic criteria. Limited information on previous treatment. Fairly low dosage AMI
Bowen (1978)	50	Primary depression (Feighner criteria)	2 weeks AMI, IMI or DMI $\geq$ 150 mg/day; combination with CDZ 30–60 mg/day	TCA > COMB	Duration of study very short. Three different antidepressants. Hardly information on previous treatment
Feighner <i>et al.</i> (1979)	27 9	Primary depression (Feighner criteria), HRSD $\geq$ 20	4 weeks AMI 75–150 mg/ day; combination with CDZ 30–60 mg/day	At 2 weeks: COMB $\geq$ AMI At 4 weeks: COMB $\geq$ AMI	Dosage COMB > dosage AMI. Limited information on previous treatment. Higher drop-out rate in AMI sample (43% vs 25%)
Johnstone <i>et al.</i> (1980)	24 0	Neurotic anxiety or depression	4 weeks AMI 150 mg/day; combination with DZP 20 mg/day	At 2 weeks: COMB > AMI At 4 weeks: AMI > COMB	No diagnostic criteria. Exclusion: recently treated patients and endogenous depressives
Feet <i>et al.</i> (1985)	63	Primary depression (Feighner criteria)	8 weeks IMI 100–200 mg/ day + PLA; IMI + DZP 10 mg/day	At 4 weeks: IMI = PLA > IMI + DZP At 8 weeks: IMI + PLA = IMI + DZP	No information on previous treatment. Exclusion: patients with agitation. High IMI plasma levels in COMB sample
Fawcett <i>et al.</i> (1987)	79	Major depression DSM-III, HRSD $\geq$ 20	DMI mean 230 mg/day; combination with ALP mean 4.6 mg/day	DMI = COMB	Exclusion: psychotic depression, suicidality. No information on previous treatment. % responders not reported

See Tables I and IIa for abbreviations.

> significantly more effective ($p < 0.005$); $\geq$ trend towards greater efficacy ($p < 0.1$); = no statistically significant difference.

* result is always measured with the main outcome criterion.

cerning previous treatment is very limited or even absent, while in only four studies was an adequate dose of TCA used (Bowen, 1978; Johnstone *et al.*, 1980; Feet *et al.*, 1985; Fawcett *et al.*, 1987). Monotherapy with a TCA is found to be significantly superior in one study (Bowen, 1978), but only as measured by a self-rating scale (the Beck Depression Inventory). In two studies a trend in favour of monotherapy was found (Jacobson, 1978; Johnstone *et al.*, 1980). In the remaining six studies no differences between the groups were found, although in one study (Rickels *et al.*, 1970) combination therapy had a superior effect in a subgroup of patients with a high level of anxiety. Thus, the assumption that BZs would reduce the efficacy of TCAs when used simultaneously (Bowen, 1978; Lipman *et al.*, 1986) cannot be either rejected or

confirmed. Although no differences were found at the end of treatment Week 4, in three studies combination therapy appears superior early in the treatment course, at Week 2 (Hare, 1971; Feighner *et al.*, 1979; Johnstone *et al.*, 1980). This implies that combination therapy with a TCA and a CBZ is justified for the first 2 weeks of treatment. As the efficacy of combination therapy over a period longer than 4–6 weeks has not been studied, the hypothesis that the prolonged use of BZ would lead to chronicity of the depression (Weissmann and Klerman, 1977; Johnson, 1985) remains to be examined.

Can benzodiazepines worsen depression?

Ryan *et al.* (1968) have reported on patients treated with diazepam experiencing an increase in depressive symp-

toms. Weissman and Klerman (1977) have reported that the use of BZs can be associated with a chronic course of depression. However, only two (Lipman *et al.*, 1986; Johnstone *et al.*, 1980) out of the 18 placebo-controlled studies reviewed here report placebo being superior over a BZ on mood and hostility. In nine studies the BZ was superior over placebo, while in seven studies no statistically significant difference was found. Lydiard *et al.* (1987) found that 15 of 46 panic disorder patients treated with alprazolam developed a major depression. The emergence of depression has also been observed during maintenance treatment with alprazolam in patients with anxiety disorders. These depressive symptoms were treated by adding a TCA or lowering the dose of the BZ. It is not surprising that some patients with anxiety disorders develop depressive symptoms, because the occurrence of major depression in these patients is quite common, particularly in panic disorder patients (Grunhaus, 1988). Whether the BZ did not prevent the emergence of spontaneous depression or whether the BZ provoked the depressive symptoms in patients with anxiety disorders cannot be concluded.

DISCUSSION

In evaluating the efficacy of BZs as antidepressants, CBZs and TBZs are considered separately. The reason is that in almost all recent studies a TBZ is investigated, whereas only a few include a CBZ. Most studies on CBZs were performed in the 1970s. Both types of BZ are compared with each other in a few studies, TBZs showing clear superiority over CBZs. This may be due to TBZs having antidepressant effects in contrast to CBZs. On the other hand, the difference may be due to TBZs having been tested in higher doses. Studies comparing CBZs with placebo and/or TCAs in major depression do not produce clues for an antidepressant effect, i.e. an effect on the core symptoms of depression (anhedonia, psychomotor retardation and anergia). The doses investigated are similar to the doses for anxiolytic treatment. Since TBZs appear to be effective in doses approximately twice as high as the recommended dose for anxiolytic treatment, it cannot be excluded that CBZs also would be effective in the higher dose range. CBZs do have an effect on some symptoms in depressed patients, such as sleep disturbances and anxiety. These effects, however, could be ascribed to general anxiolytic properties of the CBZs and they do not indicate an antidepressant effect proper. CBZs show some efficacy in populations that were classified in retrospect as minor depression, although CBZs appear to be inferior to TCAs. In most of these studies patients carry a diagnosis of "neurotic depressive reaction", "psychoneurotic reaction with depression" or "anxious depression". It is unclear whether these patients are suffering from depression, anxiety dis-

orders, or both. The efficacy of CBZs in these studies could be based on the response of patients suffering from anxiety disorders rather than depressive disorders. In several studies carried out in in-patients, TBZs have been found to be effective in mild to moderate major depression. In the majority of the studies in which TBZs were superior over placebo a threshold 21-item HAM-D score of 18 was used for inclusion, which indicates that patients probably were not severely depressed. In most of these studies patients with bipolar depression, predominant psychomotor retardation or psychotic features were excluded. TBZs turned out to be inferior to TCAs in studies involving exclusively in-patients, patients with melancholic depression or patients with severe major depression. Furthermore, when considering the studies in which TBZs were superior over placebo, the positive effects were mainly on anxiety and sleep. With the exception of psychomotor retardation, TBZs showed no effect on core symptoms of depression, such as anergia, weight loss and diurnality. As discussed above, the dose of the TBZ alprazolam used is approximately twice as high as the recommended dose for the treatment of anxiety disorders. It is concluded that TBZs are effective only for patients with mild to moderate major depression and less so for patients with more characteristic endogenous or melancholic depression. An important issue is whether patients responsive to TBZs just suffer from mild to moderate major depression or whether they suffer from a different form of major depression. It has been suggested, for instance, that some depressions are secondary to an anxiety disorder (Cloninger *et al.*, 1990; Frances *et al.*, 1992). At the symptom level primary depression may be best discriminated from secondary depression by the presence of distinct quality of mood, diminished interest and psychomotor retardation (Robinson *et al.*, 1990). This is very similar to what are called the core symptoms of depression (Schatzberg and Cole, 1978; Johnson, 1985). It is not clear, therefore, whether effects on these core symptoms should be called an antidepressant effect proper—as we do in this paper—or whether it should be called an effect on primary depression. In the latter case TCAs would be effective in both primary and secondary depression, while CBZs would be effective in secondary depression only. Similarly, BZs could be supposed to have a beneficial effect in patients with "mixed anxiety depression" as defined in the ICD-10, as opposed to endogenous or melancholic depression. Treatment with both a TCA and a BZ (combination therapy) has been studied only over a period of 4–6 weeks. BZs may contribute to the antidepressant response in the first 2 weeks because of a faster onset of action and because of effects on symptoms such as anxiety and sleep disturbances. Beyond the first weeks of treatment neither positive nor negative effects of BZs in combination with TCAs were

ascertained, which makes combination therapy beyond this period not indicated.

In conclusion, no antidepressant effect proper for CBZs in doses normally used for anxiolysis is apparent from the double-blind random assignment studies discussed. TBZs—mostly alprazolam—are effective in patients with mild to moderate major depression in doses roughly twice those used for anxiolysis, but it is questionable whether the core symptoms of depression ameliorate and TBZs are clearly inferior to TCAs in endogenous or melancholic depression.

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